

```
Processando config 2, seed 4, teste: BlockAlphabit
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 3.78
p-value of test : 0.15
```

```
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CPU time used : 00:00:00.03
```

Generator state:

16777216 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 4.70
p-value of test : 0.79

CPU time used : 00:00:00.01

Generator state:

33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 120.88
p-value of test : 0.66

CPU time used : 00:00:00.03

Generator state:

50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33041.26
p-value of test : 0.14

CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 147.63
p-value of test : 0.95

CPU time used : 00:00:00.01

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 320.44
p-value of test : 0.26

CPU time used : 00:00:00.01

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.083
p-value of test : 0.53

CPU time used : 00:00:00.03

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	42.57
p-value of test	:	0.10

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	32.83
p-value of test	:	0.38

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	31.40
p-value of test	:	0.50

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	22.63
p-value of test	:	0.60

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	8.24
p-value of test	:	0.91

CPU time used : 00:00:00.07

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	183.16	
p-value of test	:	6.7e-14	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	128.69	
p-value of test	:	1.4e-6	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	191.10	
p-value of test	:	0.05	

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	53.23	
p-value of test	:	0.62	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	29.92	
p-value of test	:	0.57	

CPU time used : 00:00:00.05

Generator state:
150994944 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.32
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):

(eps1 means a value < 1.0e-15):

Test	p-value
9 RandomWalk1 H (L = 320)	6.7e-14
9 RandomWalk1 M (L = 320)	1.4e-6

All other tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
      Starting Alphabit:  nb = 16777216
      Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 3.18
p-value of test : 0.20

CPU time used : 00:00:00.01

Generator state:

167772160 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 5.49
p-value of test : 0.70

CPU time used : 00:00:00.01

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 131.82
p-value of test : 0.39

CPU time used : 00:00:00.01

Generator state:
201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32751.03
p-value of test : 0.53

CPU time used : 00:00:00.03

Generator state:
218103808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 174.40
p-value of test : 0.54

CPU time used : 00:00:00.03

Generator state:
234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 319.00
p-value of test : 0.28

CPU time used : 00:00:00.01

Generator state:
251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.59
p-value of test : 0.28

CPU time used : 00:00:00.01

Generator state:
268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	42.78
p-value of test	:	0.10

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	40.68
p-value of test	:	0.11

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	29.07
p-value of test	:	0.62

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	23.50
p-value of test	:	0.55

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	16.72
p-value of test	:	0.34

CPU time used : 00:00:00.07

Generator state:

285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	483.14	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	297.65	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	219.92	
p-value of test	:	1.2e-3	

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	66.04	
p-value of test	:	0.19	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	34.49	
p-value of test	:	0.35	

CPU time used : 00:00:00.05

Generator state:
301989888 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.26
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value $< 1.0e-300$):
(eps1 means a value $< 1.0e-15$):

Test	p-value
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps

All other tests were passed

XX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1 , we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = $2.9802323e-08$, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	1.43
p-value of test	:	0.49

CPU time used : 00:00:00.03

Generator state:
318767104 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 11.07
p-value of test : 0.20

CPU time used : 00:00:00.01

Generator state:

335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 117.99
p-value of test : 0.73

CPU time used : 00:00:00.01

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32578.99
p-value of test : 0.77

CPU time used : 00:00:00.03

Generator state:
369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 205.47
p-value of test : 0.07

CPU time used : 00:00:00.01

Generator state:
385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 3073.55
p-value of test : eps *****

CPU time used : 00:00:00.01

Generator state:
402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.082
p-value of test : 0.47

CPU time used : 00:00:00.01

Generator state:
419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 1411.18
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 448.30
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 137.38
p-value of test : 5.1e-15 *****

Test on the values of the Statistic R

Number of degrees of freedom : 25
ChiSquare statistic : 16.27
p-value of test : 0.91

Test on the values of the Statistic C

Number of degrees of freedom : 15
ChiSquare statistic : 11.40
p-value of test : 0.72

CPU time used : 00:00:00.07

Generator state:
436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	534.64	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	317.16	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	222.78	
p-value of test	:	7.6e-4	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	59.82	
p-value of test	:	0.37	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	22.28	
p-value of test	:	0.90	

CPU time used : 00:00:00.05

Generator state:
452984576 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.26
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
6 HammingIndep, L = 32	eps
8 RandomWalk1 H (L = 64)	eps
8 RandomWalk1 M (L = 64)	eps
8 RandomWalk1 J (L = 64)	5.1e-15
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	7.6e-4
-----	-----

All other tests were passed

XX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 2.19
p-value of test : 0.33

CPU time used : 00:00:00.01

Generator state:
469761792 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 11.54
p-value of test : 0.17

CPU time used : 00:00:00.01

Generator state:
486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 123.57
p-value of test : 0.59

CPU time used : 00:00:00.01

Generator state:

503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33583.42
p-value of test : 7.8e-4 *****

CPU time used : 00:00:00.03

Generator state:
520093440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	177	
Chi-square statistic	:	6169.92	
p-value of test	:	eps	*****

CPU time used : 00:00:00.01

Generator state:
536870656 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305	
Chi-square statistic	:	3075.18	
p-value of test	:	eps	*****

CPU time used : 00:00:00.01

Generator state:
553647872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.024
p-value of test : 0.51

CPU time used : 00:00:00.01

Generator state:
570425088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 1818.69
p-value of test : eps *****

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 811.06
p-value of test : eps *****

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 296.13
p-value of test : eps *****

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	11.08
p-value of test	:	0.9926

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	15.51
p-value of test	:	0.42

CPU time used : 00:00:00.07

Generator state:
587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	978.74
p-value of test	:	eps *****

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	444.83
p-value of test	:	eps *****

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	315.72
p-value of test	:	2.8e-12 *****

Test on the values of the Statistic R

Number of degrees of freedom : 57
ChiSquare statistic : 75.84
p-value of test : 0.05

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 40.92
p-value of test : 0.13

CPU time used : 00:00:00.05

Generator state:
603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
4 MultinomialBitsOver, L = 16	7.8e-4
5 HammingIndep, L = 16	eps
6 HammingIndep, L = 32	eps
8 RandomWalk1 H (L = 64)	eps
8 RandomWalk1 M (L = 64)	eps
8 RandomWalk1 J (L = 64)	eps
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	2.8e-12

All other tests were passed

XX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 2.68
p-value of test : 0.26

CPU time used : 00:00:00.01

Generator state:

620756480 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 8.35
p-value of test : 0.40

CPU time used : 00:00:00.01

Generator state:
637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 154.74
p-value of test : 0.05

CPU time used : 00:00:00.01

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	: 32768
Value of the statistic	: 32982.93
p-value of test	: 0.20

CPU time used : 00:00:00.03

Generator state:

671088128 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 173.99
p-value of test : 0.55

CPU time used : 00:00:00.01

Generator state:

687865344 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 269.72  
p-value of test                   : 0.93  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                  : 1.25  
p-value of test                   : 0.11  
-----
```

```
-----  
CPU time used                     : 00:00:00.01  
-----
```

Generator state:

721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64  
-----
```

Test on the values of the Statistic H

```
-----  
Number of degrees of freedom      : 32  
ChiSquare statistic               : 32.38  
p-value of test                   : 0.45  
-----
```

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	33.88
p-value of test	:	0.33

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	31.83
p-value of test	:	0.48

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	20.89
p-value of test	:	0.70

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	16.00
p-value of test	:	0.38

CPU time used : 00:00:00.07

Generator state:

738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	1087.71
p-value of test	:	eps *****

```

-----
Test on the values of the Statistic M

Number of degrees of freedom      :    62
ChiSquare statistic               :   662.03
p-value of test                   :    eps      *****

```

```

-----
Test on the values of the Statistic J

Number of degrees of freedom      :   160
ChiSquare statistic               :   423.48
p-value of test                   :    eps      *****

```

```

-----
Test on the values of the Statistic R

Number of degrees of freedom      :    57
ChiSquare statistic               :    35.74
p-value of test                   :     0.99

```

```

-----
Test on the values of the Statistic C

Number of degrees of freedom      :    32
ChiSquare statistic               :    37.14
p-value of test                   :     0.24

```

```

-----
CPU time used                     :   00:00:00.05

```

```

Generator state:
 754973952 bits have been read.

```

===== Summary results of Alhabbit =====

```

Version:           TestU01 1.2.3
Generator:         ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:    16777216
Number of statistics: 17
Total CPU time:    00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

```

Test	p-value
-----	-----
9 RandomWalk1 H (L = 320)	eps
9 RandomWalk1 M (L = 320)	eps
9 RandomWalk1 J (L = 320)	eps
-----	-----

All other tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 32
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 0.41
p-value of test                    : 0.81
```

```
-----
CPU time used                      : 00:00:00.01
```

```
Generator state:
```

```
771751168 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 32
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
```

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 6.55
p-value of test : 0.59

CPU time used : 00:00:00.03

Generator state:

788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 143.58
p-value of test : 0.16

CPU time used : 00:00:00.01

Generator state:

805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32737.30
p-value of test : 0.55

CPU time used : 00:00:00.03

Generator state:

822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 183.28
p-value of test : 0.36

CPU time used : 00:00:00.01

Generator state:

838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 314.44
p-value of test : 0.34

CPU time used : 00:00:00.01

Generator state:

855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.88
p-value of test : 0.81

CPU time used : 00:00:00.01

Generator state:

872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	21.11
p-value of test	:	0.93

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	20.96
p-value of test	:	0.91

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	23.41
p-value of test	:	0.87

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	17.09
p-value of test	:	0.88

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	8.84
p-value of test	:	0.89

CPU time used : 00:00:00.07

Generator state:

889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config2_seed4.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62	
ChiSquare statistic	:	1372.65	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	62	
ChiSquare statistic	:	743.08	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	160	
ChiSquare statistic	:	348.35	
p-value of test	:	5.6e-16	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	57	
ChiSquare statistic	:	45.55	
p-value of test	:	0.86	

Test on the values of the Statistic C

Number of degrees of freedom	:	32	
ChiSquare statistic	:	41.24	
p-value of test	:	0.13	

CPU time used : 00:00:00.05

Generator state:
905968640 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.23
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

	Test	p-value
9	RandomWalk1 H (L = 320)	eps
9	RandomWalk1 M (L = 320)	eps
9	RandomWalk1 J (L = 320)	5.6e-16

All other tests were passed